Sameer Choudhary

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Education

International Institute of Information Technology (IIIT), Naya Raipur Aug 2024 – May 2028
B.Tech in Data Science & Artificial Intelligence; CGPA: 8.92/10 *Raipur, India*

Relevant Coursework

AI Specialization: Machine Learning, Deep Learning, Natural Language Processing, Agentic AI, Computer Vision, Reinforcement Learning, Representation Learning
CS Core: Data Structures & Algorithms, Database Management Systems, Operating Systems

Projects

bhasha-shield: Multilingual Red-Teaming of LLM Agents | *vLLM, AgentDojo* | GitHub Sep 2026 – Present

- Built a harness that checks if **LLM agents** can be hijacked by hidden instructions in tool outputs (**prompt injection**) in English, Hindi, Romanized Hindi, Bengali and code-mixed text, on Qwen2.5-14B (4-bit AWQ, **vLLM**)
- Benchmarked **7 defenses**: a zero-cost "sandwich" prompt cut attack success from **20.5% to 6.8%** ($p = 0.004$, $n = 88$) without hurting task completion; a DeBERTa guard cut completion from 46% to 22.7%
- Found English-tuned safety classifiers fail on Indian languages: ProtectAI flagged **47%** of harmless Hindi tool output (2% in English); recalibrating PromptGuard 2 for Hindi raised detection from 0% to **51%** at 2% FPR

RecoverAI: LLM Debt-Recovery Agent | *FastAPI, Gemini, MongoDB, Redis* | GitHub Apr 2026 – Jun 2026

- Built an **LLM agent** that automates invoice collection for small businesses: **Gemini** extracts invoice data from emailed PDFs, then the agent runs scheduled WhatsApp/email follow-ups and drafts MSMED Act legal notices
- Classified Hindi/Hinglish debtor replies (promise-to-pay, dispute, stall) to pick the agent's next action; **90% accuracy** on an 80-message labelled test set, vs 54% for a keyword baseline
- Engineered the **FastAPI** + MongoDB backend with background workers, multi-model LLM fallback, JWT and Google OAuth login, **Redis** rate limiting and SSRF guards; deployed on Vercel

Kangri–Hindi Neural Machine Translation (team of 2) | *PyTorch, Transformers, ONNX* Jul 2025 – Present

- Fully fine-tuned **M2M-100 (418M)** and **NLLB-200 (600M)** on **23,716** cleaned pairs for endangered Kangri; reached **39.14 BLEU** Kangri→Hindi vs 28.17, the best previously reported (Bisht & Gupta, different test split)
- Showed task-specific fine-tuning beats prompting: NLLB-200 tuned on just 4,000 pairs (28.07 BLEU) outperformed **5-shot GPT-4o** (24.32) and Llama 3.3 70B (15.19)
- Tested 3 meta-learning methods (FOMAML, Reptile, ANIL) on A100 GPUs; a meta-learning warm-up before fine-tuning added **+2.0 to +3.3 BLEU**, equal to doubling the training data
- Wrote a TF-IDF data-leakage audit that caught a model memorising test sentences; exported models to **ONNX**

Sentinel: Real-Time Road Accident Detection | *YOLOv8, OpenCV, FastAPI, Docker* | GitHub Jan 2026

- Fine-tuned **YOLOv8m** on a custom accident dataset with normal-traffic negatives to cut false alarms; reached **84.1% precision**, 69.1% recall and **78.0% mAP@50**
- Shipped a FastAPI + WebSocket backend and Next.js dashboard that streams detections live and alerts only after 5 consecutive positive checks to avoid false alarms; runs on CPU (about 330 ms per inference) in Docker

Also built: BlogGraph (LangGraph agent that writes technical blogs grounded in live web search, i.e. RAG), ShowYourWork (MCP server keeping a signed, tamper-evident log of AI-assisted writing; 46 tests), AutoElicitate (auto-prepares any CSV for ML; an LLM fixes the pipeline when it crashes)

Technical Skills

Languages: Python, C++, SQL, Java, TypeScript, JavaScript
ML / DL: PyTorch, Hugging Face Transformers, scikit-learn, Pandas, NumPy, YOLOv8, OpenCV
LLMs & GenAI: LLM agents, RAG, vector DBs, LoRA/PEFT, LangGraph, MCP, vLLM, AWQ quantization, guardrails
Backend & MLOps: FastAPI, WebSockets, Docker, ONNX, MongoDB, Redis, Git, Hugging Face Spaces, Vercel
Evaluation: Precision/recall/F1, mAP, AUC, BLEU, COMET, data-leakage checks, significance testing

Achievements

Solved **700+** data structures and algorithms problems on **LeetCode** and **Codeforces**